

Diagnostic performance of an artificial intelligence based software for opportunistic osteoporosis detection using chest CT

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ABSTRACT

Objective: To investigate the diagnostic accuracy of artificial intelligence (AI) derived bone mineral density (BMD) for osteoporosis screening on chest CT, using dual-energy X-ray absorptiometry (DXA) as the reference. **Methods:** Subjects who underwent chest CT for lung cancer screening with T11 vertebra fully included during July 2023 to December 2023 were included retrospectively. Regions of interest (ROIs) were automatically placed on T11, T12 and L1. BMD was calculated using an AI-based software. Diagnostic accuracy of the AI-derived BMD of T11 was evaluated by receiver operating characteristic (ROC) curves using DXA as the reference. In a subgroup of subjects with the three vertebrae fully included, the AUCs using BMD of T12, L1 or a mean of multiple vertebrae were compared with those using T11 BMD alone.

Results: A total of 560 subjects (49.4 ± 9.2 years, 312 men) were included. The AUC of the AI-derived BMD of T11 for osteoporosis detection was 0.898 (0.869, 0.927), for detecting abnormal BMD was 0.834 (0.800, 0.869). The cut-off values for achieving a sensitivity of 90 % for osteoporosis and abnormal BMD diagnosis were 144 mg/cm³ and 175 mg/cm³, respectively. No differences were shown between the AUCs using BMD of T11 alone vs. using BMD of T12, L1, or the mean of two or three vertebrae (for osteoporosis detection, AUCs, 0.891–0.900, p values, 0.234–0.921, for abnormal BMD detection, AUCs, 0.833–0.840, p values, 0.343–0.888).

Conclusion: The AI-derived BMD of T11 on chest CT scans has a high diagnostic accuracy for opportunistic detection of osteoporosis and abnormal BMD using DXA as the reference.

Note: This retrospective study was approved by the research ethical committee of Southern University of Science and Technology Hospital (No.016, 2024).

Introduction

Osteoporosis is a systemic skeletal disease, defined as a decrease of bone mass and deterioration of bone tissue microarchitecture [1]. It is related to an increase in fractures of multiple skeletal sites, which can lead to an increase in morbidity and mortality in the elderly [1]. Bone mineral density (BMD) measurement on imaging is important in the diagnosis of osteoporosis according to the diagnostic criteria from WHO, which defined osteoporosis and osteopenia using T-scores of BMD acquired by dual-energy X-ray absorptiometry (DXA) [2]. Screening of osteoporosis is recommended for all postmenopausal women and subjects aged 50 years or older [3]. As the chest CT scan is nowadays widely used for lung cancer screening, evaluation of the vertebral BMD on chest

CT scans would be beneficial for osteoporosis screening at the same time [4–8]. Direct Hounsfield units (HU) value measurements on chest CT scans for opportunistic osteoporosis screening have been evaluated by several researchers [7–10]. Recently a measurement of the mean HU values of all thoracic vertebrae aided by automatic regions of interest (ROI) placements yielded an area under receiver operating curve (AUC) of 0.831 and 0.972 for detection of osteopenia and osteoporosis, using DXA results as the reference standard [7]. However, different cut-off values have been derived from various studies [9], which could be a practical problem for HU value measurement to be used in clinical practice.

Opportunistic osteoporosis screening on chest CT scans has been enhanced by artificial intelligence (AI) technologies [11]. Algorithms for

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BMD estimation using chest CT images has been trained using QCT as the reference [4,12–14]. The AI-derived BMD values have a high consistency with real QCT BMD values in the test datasets, with a highest reported AUC of 0.970 for osteoporosis detection [12]. As DXA is the recommended method for osteoporosis screening [1,15] and also the only method that has been used in all osteoporosis treatment trials [3], the diagnostic accuracy of the AI-derived BMD for osteopenia and osteoporosis detection using DXA as the reference standard should be evaluated, which has not yet been performed in literature. This would be important and useful for radiologists and clinicians to interpret the AI-derived BMD values in daily patient care.

In the current study, we aimed to investigate the diagnostic accuracy of an AI based software previously trained to generate QCT-like BMD results for opportunistic osteopenia and osteoporosis detection on chest CT scans, using DXA results as the reference standard.

Materials and methods

This retrospective study was approved by the research ethical committee of Southern University of Science and Technology Hospital (No.016, 2024). Informed consent was waived.

Study population

A total of 1709 subjects aged 18–90 years old who came to our institution for routine health check and underwent unenhanced chest CT scans during the period of July 2023 to Dec 2023 were reviewed from the Picture Archiving and Communication System (PACS) system. Exclusion criterion were: (1) without DXA results acquired on the same day ($n = 1138$); (2) metal implants in the spine or in the left hip ($n = 3$); (3) calculation failed by the AI module ($n = 2$); (4) T11 vertebra was not fully included on the CT images ($n = 2$); (5) presence of osteolytic/osteoblastic lesions or vertebral fractures in the T11 vertebra ($n = 4$). In total, 560 subjects were included in this study.

Subject classification

The subjects were classified into normal, osteopenia and osteoporosis according to the lowest T value of total lumbar, femoral neck and total hip. For all subjects, osteoporosis was defined as a T-score ≤ -2.5 , osteopenia was defined as $-2.5 < \text{T-score} < -1.0$, and normal BMD was defined as a T-score ≥ -1.0 . The subgroup of subjects with osteoporosis or osteopenia was referred to as subjects with abnormal BMD in this study.

Image acquisition

DXA and CT scans

DXA scans were performed on a Hologic device (Discovery WI, Hologic Inc. USA). The ROI of BMD measurements included the 1st to 4th lumbar vertebrae, left femoral neck and left total hip.

Chest CT scans were performed on a 256-slice CT scanner (Revolution, GE healthcare, Milwaukee, USA) and a 64-slice CT scanner (Optima CT680, GE Healthcare, Milwaukee, USA) using a tube voltage of 120 kV and an automatic tube current. Thin slice axial images (slice thickness: 0.625 mm) were reconstructed for BMD calculation.

Automatic AI-derived BMD analysis

AI derived QCT-like BMD analysis was automatically carried out using Phantom-less Bone Mineral Density Calculation Software (uAI-Discover PLBMD, version R001, United Imaging, Shanghai, China). The AI software automatically placed circular ROIs on T11, T12 and L1 (when available on the images) vertebrae and automatically estimate the BMD values within the ROIs. The phantomless software was developed using the Python programming language. By incorporating data from multiple clinical centers and a range of CT equipment, the

developers chose reference substances with distinct attenuation that exhibit stable CT HU values across the general population, ensuring their reliability as internal reference materials. By employing deep learning techniques and machine learning algorithms, the system ensures precise BMD measurements. It effectively compensates for variations in scanning devices and factors affecting CT values, such as beam hardening, thereby facilitating the accurate conversion of CT values into reliable BMD measurements. The system integrates data from internal reference tissues, ensuring compatibility with CT images from diverse CT models, and is characterized by rapid computation speed, high efficiency, and robust performance under varying conditions. ROI selection and BMD measurements using the AI software are shown in Fig. 1.

Statistical analysis

Normally distributed variables were described as mean $\pm$ standard deviation (SD). In a subgroup of subjects with all T11, T12 and L1 BMD values available, correlation between the AI-estimated BMD values (of T11, T12, L1, the average of T11 and T12, the average of T11, T12 and L1) and the T-scores (of the total lumbar spine, femoral neck and total hip) was analyzed using Pearson correlation. The Pearson correlation coefficients from different vertebrae were compared. For further analysis of diagnostic accuracy, the AI-estimated BMD value of T11, T12 and L1 was used and compared. Receiver operating characteristic curve (ROC) analysis of the AI-estimated BMD was performed using DXA

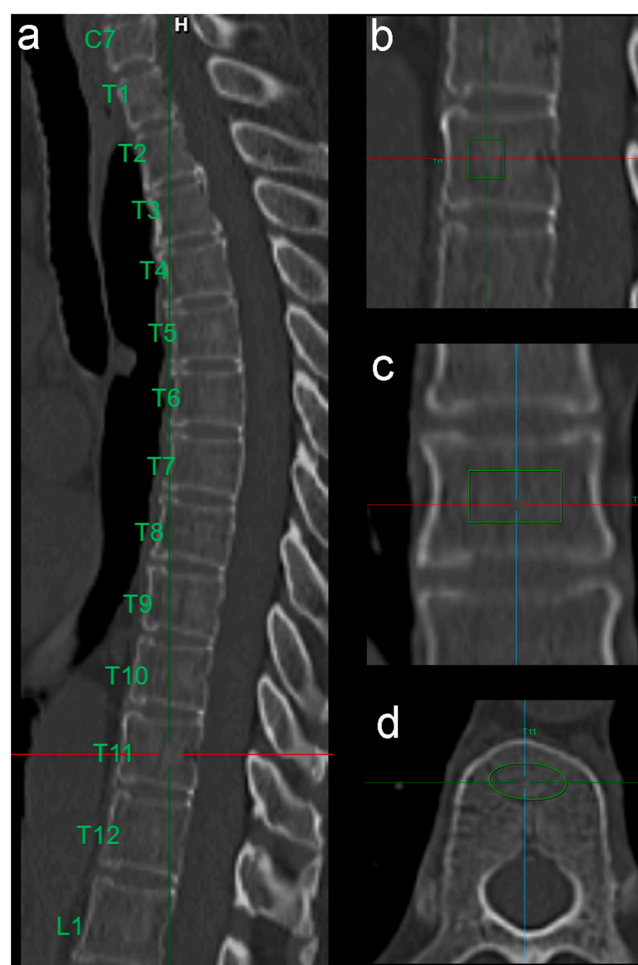


Fig. 1. Regions of interest (ROI) selection and bone mineral density (BMD) measurements using the AI software. The T11, T12 and L1 vertebrae are automatically located by the AI software and ROIs (shown as green circles and rectangles) were automatically placed.

results as the reference standard. The optimal cut-off values achieving a highest Youden's Index and cut-off values achieving a sensitivity of 90 % or a specificity of 90 % were derived. The sensitivity, specificity and AUCs were calculated and compared. AUCs for osteoporosis/osteopenia detection using T11 BMD alone were compared with the AUCs derived by using T12 BMD alone, L1 BMD alone, the mean of T11 and T12 BMD, and the mean of T11, T12 and L1 BMD, respectively. For the AI-estimated BMD values, the diagnostic accuracy of the existed American College of Radiology (ACR) QCT diagnostic criteria was also tested, which defined osteoporosis and osteopenia [5] as BMD < 80 mg/cm³ and 80 to 120 mg/cm³, respectively. Statistical analysis was performed using SPSS (version 25.0, IBMcorp, Armonk, N.Y., USA) and MedCalc (version 20.1, Medcalc Software Ltd, Ostend, Belgium).

Results

In total, 560 subjects (age range, 25–80 years, mean age, 49.4 ± 9.2 years, 312 men) were included in this study. The chest CT scan and DXA of the same subject were all performed on the same day. The demographics and DXA results of all subjects and subgroups were summarized in Table 1.

Correlation between AI-derived BMD of different vertebral levels and DXA results

A subgroup of 410 subjects (mean age, 49.3 ± 8.9 years, age range, 31–80 years, 223 men) had T11, T12 and L1 BMD values available. The AI-derived BMD of T11, T12 and L1 and the correlation coefficients between the AI-derived BMD and DXA results are summarized in Supplementary table 1. For both lumbar and hip, no differences were shown between the correlation coefficients derived from the BMD values at different vertebra/ the average value of vertebrae. In all 410 subjects, the mean BMD value of T11 was significantly higher than the mean BMD value of T12 (150 mg/cm³ vs 144 mg/cm³, $p < 0.001$) and the mean BMD value of L1 (150 mg/cm³ vs 141 mg/cm³, $p < 0.001$). The correlation coefficient between T11 BMD value and L1 BMD value was 0.944 (95 % confidence interval (CI), 0.933, 0.954, $p < 0.001$).

Table 1
Demographics and DXA results of the study population.

Variable*	Male (n = 312)	Female (n = 248)	Total (n = 560)
Age (years)	48.8 ± 9.1 (25, 80)	50.2 ± 9.3 (30, 73)	49.4 ± 9.2 (25, 80)
Height (cm)	170.8 ± 6.7 (150, 190)	159.1 ± 5.3 (141, 178)	165.6 ± 8.5 (141, 190)
Weight (kg)	73.7 ± 10.8 (46, 124)	57.6 ± 8.0 (38, 86)	66.6 ± 12.5 (38, 124)
Body mass index (kg/m ²)	25.2 ± 2.9 (17.5, 35.5)	22.8 ± 3.1 (16.6, 33.3)	24.1 ± 3.2 (16.6, 35.5)
Number of subjects with T12 fully included	288 (92.3 %)	232 (93.5 %)	520 (92.8 %)
Number of subjects with L1 fully included	226 (72.4 %)	189 (76.2 %)	415 (74.1 %)
Total lumbar T-score	−1.01 ± 1.39 (−4.4, 3.0)	−0.96 ± 1.39 (−4.8, 4.0)	−0.99 ± 1.34 (−4.8, 4.0)
Femoral neck T-score	−1.01 ± 1.09 (−9.0, 3.7)	−1.14 ± 1.05 (−3.8, 2.1)	−1.07 ± 1.07 (−9.0, 3.7)
Total hip T-score	−0.66 ± 0.92 (−3.2, 2.8)	−0.69 ± 1.02 (−3.2, 2.3)	−0.67 ± 0.96 (−3.2, 2.8)
Number of osteopenia#	135 (43.2 %)	109 (43.9 %)	244 (43.6 %)
Number of osteoporosis#	71 (22.7 %)	50 (20.1 %)	121 (21.6 %)

* Data are mean ± standard deviation, with range in parentheses. #Osteopenia and osteoporosis are classified according to the lowest T-value of total lumbar, femoral neck and total hip measured by DXA, with percentages in parentheses.

ROC analysis of the AI-derived BMD using the lowest DXA T-value as the reference standard

AUC values and results of ROC analysis in the subgroup subjects with T11, T12 and L1 BMD values available were provided in Table 2 and Supplementary Table 2. Using the AI-derived BMD value of T11 vertebra alone for detection of osteoporosis or abnormal BMD, the AUCs were not different from those using T12 alone, L1 alone, the mean value of T11 and T12, and the mean value of T11, T12 and L1. To be noted, to achieve a 90 % sensitivity for diagnosis of osteoporosis/abnormal BMD, the cut-off value of using BMD of T11 vertebra alone was higher than the cut-off value of using BMD of T12, L1 or the mean value of T11, T12 and L1.

ROC curves for detection of osteoporosis and abnormal BMD using the AI-derived BMD of T11 are shown in Fig. 2. AUC values and results of ROC analysis are summarized in Table 3. In all included subjects, the AUC for detection of abnormal BMD including osteopenia and osteoporosis was 0.834 (0.800, 0.869), for detection of osteoporosis was 0.898 (0.869, 0.927). Using a BMD value < 144 mg/cm³ and a BMD value < 175 mg/cm³ as the cut-off values could achieve a sensitivity of 90 % for detection of osteoporosis and abnormal BMD, respectively. Applying these cut-off values in all female subjects yielded a sensitivity of 94 % and a specificity of 70 % for detection of osteoporosis, and a sensitivity of 87 % and a specificity of 64 % for detection of abnormal BMD. In male subjects, the same cut-off values had a sensitivity of 87 % and a specificity of 64 % for detection of osteoporosis, and a sensitivity of 93 % and a specificity of 47 % for detection of abnormal BMD.

Comparison between the diagnosis rate of osteoporosis and osteopenia of DXA and AI-derived BMD of T11 vertebra using the ACR QCT diagnostic criteria

According to the ACR QCT diagnostic criteria, 425 subjects (75.8 %) were normal, 114 subjects (20.3 %) were osteopenia and 21 subjects (3.7 %) were osteoporosis using the AI-derived BMD values of T11 vertebra. The contingency table between the AI-derived BMD classification according to the ACR diagnostic criteria and the DXA results was shown in Table 4. Taking the lowest DXA T-score as the reference standard, using the ACR diagnostic criteria as cut-off values for interpretation of the AI-derived BMD results of T11 vertebra yielded a sensitivity of 35 % (128/365) and specificity of 96 % (188/195) for detection of abnormal BMD (i.e. osteoporosis and osteopenia). And the sensitivity and specificity for detection of abnormal lumbar BMD and abnormal hip BMD were 43 % (118/272), 94 % (271/288) and 57 % (117/312), 93 % (230/248) respectively.

Discussion

In the current study, using an AI-based software that automatically placed ROIs on the T11, T12, L1 vertebrae (when included in the scan) and automatically estimated the BMD values within the ROIs, we found that the AI-derived BMD value of the T11 vertebra had a high AUC for detection of osteoporosis and abnormal BMD including osteopenia and osteoporosis using the lowest DXA T-value of lumbar and hip as the reference standard. The cut-off values for achieving a sensitivity of 90 % for osteoporosis and abnormal BMD diagnosis were 144 mg/cm³ and 175 mg/cm³, respectively. Applying the ACR QCT diagnostic criteria when using the AI-derived BMD value of the T11 vertebra yielded a high specificity but an inadequate sensitivity for diagnosis of abnormal BMD. Meanwhile, no differences in the AUC values or the correlation coefficients with DXA results were shown when using the BMD of T11, T12, L1 vertebra alone or the mean value of T11, T12 or the mean value of T11, T12 and L1 vertebrae for the diagnosis of osteoporosis and abnormal BMD.

Opportunistic screening of abnormal BMD including osteopenia and osteoporosis on readily available chest or abdominal CT images are beneficial for the patients as serendipitous findings without extra

Table 2
ROC analysis of the AI-derived BMD values in 410 subjects with all T11, T12 and L1 vertebrae included, taking DXA results (the lowest T-value of lumbar and hip) as the reference standard.

Vertebra for AI-derived BMD analysis	Osteoporosis diagnosis (n = 87)			Abnormal BMD diagnosis (n = 276)			P value	
	AUC 1 (95 % CI)	Cut-off value for 90 % sensitivity, mg/cm ³	Specificity	AUC 2 (95 % CI)	Cut-off value for 90 % sensitivity, mg/cm ³	Specificity	P1	P2
T11	0.901 (0.868, 0.928)	142	71 %	0.840 (0.801, 0.874)	178	52 %	NA	NA
T12	0.891 (0.857, 0.920)	132	74 %	0.833 (0.794, 0.868)	169	54 %	0.234	0.343
L1	0.892 (0.858, 0.920)	134	70 %	0.833 (0.793, 0.868)	169	51 %	0.262	0.407
Mean of T11 and T12	0.899 (0.866, 0.927)	140	69 %	0.840 (0.801, 0.874)	173	54 %	0.681	0.872
Mean of T11, T12 and L1	0.900 (0.867, 0.927)	135	71 %	0.840 (0.801, 0.874)	170	54 %	0.921	0.888

P values are for comparison with the AUC values of T11 BMD. Abbreviations: AUC, area under the curve, CI, confidence interval, BMD, bone mineral density, AI, artificial intelligence, NA, not appropriate.

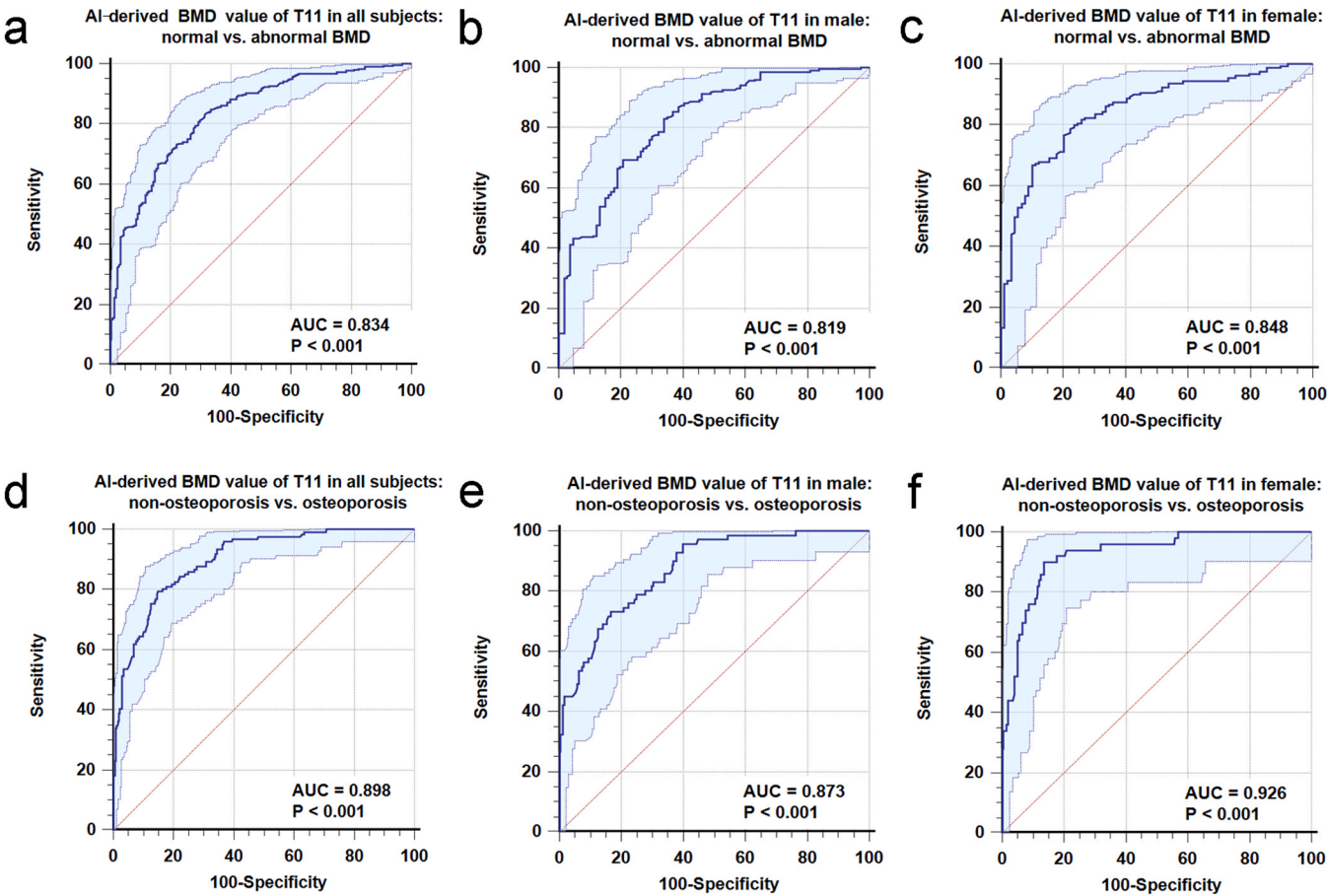


Fig. 2. ROC curves for detection of osteoporosis and abnormal BMD using the lowest DXA T-value as the reference standard. The ROC curves using the AI-derived BMD for differentiating between normal and abnormal BMD (osteopenia and osteoporosis) in all subjects (a), men (b) and women (c). (d-f) The ROC curves using the AI-derived BMD for differentiating between osteoporosis and non-osteoporosis (normal and osteopenia) in all subjects (a), men (b) and women (c).

radiation exposure. Previous studies has explored opportunistic screening of BMD abnormality using chest CT scans. QCT can be performed for BMD analysis when taking a chest CT scan [5], however, phantom is needed and the L2 vertebra needs to be included which means a larger scan range than a regular chest CT scan. Vertebral body CT attenuation value measurements on chest or abdominal CT scans for osteoporosis screening were performed using DXA results as the reference standard in several studies [7,16–18]. The reported AUC for detection of osteopenia versus normal subjects ranged from 0.74 to 0.83 using CT attenuation values of different thoracic or lumbar vertebra [7, 17,19]. To be noted, CT attenuation measurements can be inconsistent among different scanners and protocols [20]. Although measuring of CT attenuation values is simple and accessible widely across institutions,

the variability in the cut-off values for osteoporosis or osteopenia detection derived from different studies is a main limitation for its interpretation in clinical practice. Phantom-less QCT method was also developed for opportunistic osteoporosis screening using chest CT, with an in-house designed algorithm for internal calibration using CT attenuation values of fat and muscle [6]. Using the T12 vertebral BMD value calculated by the phantom-less QCT algorithm, Tang et al. reported an AUC of 0.74 for osteoporosis and for osteopenia diagnosis using the lower value of spine and hip DXA T-value as the reference standard [6]. AI-based algorithms have been trained using QCT as the reference standard to generate QCT-like BMD results for opportunistic osteoporosis screening [11,12,21]. Using QCT as the reference standard, different AI algorithms had a high accuracy in detection of osteoporosis

Table 3

ROC analysis of the AI-derived BMD values of T11 vertebra taking DXA results (the lowest T-values of lumbar and hip) as the reference standard.

Study population	Diagnosis	AUC (95 %CI)	Balanced Sen and Spe with a highest Youden Index			Threshold for achieving 90 % sensitivity (Sen)			Threshold for achieving 90 % Specificity (Spe)		
			Cut-off (mg/cm ³)	Sen	Spe	Cut-off (mg/cm ³)	Sen	Spe	Cut-off (mg/cm ³)	Sen	Spe
All (n = 560)	Osteoporosis (n = 121)	0.898 (0.869, 0.927)	126	79 %	85 %	144	90 %	66 %	118	65 %	90 %
	Abnormal BMD (n = 365)	0.834 (0.800, 0.869)	164	84 %	68 %	175	90 %	55 %	136	53 %	90 %
Women (n = 248)	Osteoporosis (n = 50)	0.926 (0.888, 0.965)	126	90 %	86 %	126	90 %	86 %	117	76 %	90 %
	Abnormal BMD (n = 159)	0.848 (0.799, 0.897)	163	77 %	80 %	180	90 %	55 %	151	67 %	90 %
Men (n = 312)	Osteoporosis (n = 71)	0.873 (0.829, 0.916)	127	73 %	83 %	145	90 %	63 %	119	58 %	90 %
	Abnormal BMD (n = 206)	0.819 (0.770, 0.868)	160	83 %	66 %	170	90 %	54 %	129	44 %	90 %

Table 4

Contingency table analysis.

DXA for reference	Diagnosis	AI-derived T11 BMD results (using ACR diagnostic criteria)			
		Normal	Osteopenia	Osteoporosis	Total
Lumbar DXA	Normal	271	17	0	288
	Osteopenia	133	45	6	184
	Osteoporosis	21	52	15	88
	Total	425	114	21	560
Hip DXA (lower T value of femoral neck and total hip)	Normal	230	17	1	248
	Osteopenia	180	77	9	266
	Osteoporosis	15	20	11	46
	Total	425	114	21	560
The lowest T value of lumbar and hip DXA	Normal	188	7	0	195
	Osteopenia	200	43	1	244
	Osteoporosis	37	64	20	121
	Total	425	114	21	560

(AUC, 0.927 to 0.987) and abnormal BMD (AUC, 0.942 to 0.984). The diagnostic performance of AI-derived QCT-like BMD results for opportunistic osteoporosis and osteopenia screening have not been evaluated using DXA as the reference standard. The agreement between QCT results and the current AI-derived BMD results has been verified elsewhere and the results were in publication, which was beyond the scope of our study. In our study, we found an AUC of 0.898 for osteoporosis diagnosis and an AUC of 0.834 for abnormal BMD diagnosis using the AI-derived BMD value of T11 vertebra, which were higher than the previously mentioned studies using CT attenuation measurements or other phantomless QCT evaluation.

In the current study, we chose the AI-derived BMD value of T11 vertebra for a detailed diagnostic performance analysis because T11 vertebra was fully included in almost all of the reviewed chest CT scans (99.6 %) while the percentage for T12 and L1 vertebra was 91 % and 73 %, respectively. As our aim is to use the AI-derived BMD on the chest CT scan for screening of osteoporosis, it is important that the evaluated vertebra can be available in the general population who had chest CT scans. Moreover, when comparing the AI-derived BMD value of T11, T12, L1 vertebra alone or the mean of the two or three vertebrae, we found that there were no evidence indicating significant differences in either the correlation with DXA results or the AUC values. These results suggested that using the AI-derived BMD value of T11 vertebra for BMD abnormality screening was as accurate as using that of T12, L1 or the mean values. This was comparable to the results from Yang et al. [7]. In their study, they found similar AUC values for osteoporosis/osteopenia diagnosis using CT attenuation values from each thoracic vertebra (from T1 to T12) and L1 vertebra. Notably, a decrease of the AI-derived BMD was shown from T11 to L1, which was also found by previous studies [7,

22,23]. On average, the AI-derived BMD of T11 was 6 mg/cm³ higher than that of T12, and 9 mg/cm³ higher than that of L1. The importance for clinical practice is that when using the AI-derived BMD of T11 vertebra for osteoporosis or abnormal BMD screening, the cut-off value should be higher than using the value from T12 or L1 vertebra.

One advantage of this study is that all subjects had the chest CT scan and DXA scan on the same day, avoiding any possible factors that may change the BMD during the time interval between CT scans and DXA scans in some other studies. Using DXA results as the reference standard, applying the ACR osteoporosis diagnostic criteria for QCT on the AI-derived BMD value of T11 had a sensitivity of 35 % and a specificity of 96 % for detection of abnormal BMD, and a sensitivity of 17 %, a specificity of 99.8 % for detection of osteoporosis. There are several possible reasons for this low sensitivity using the ACR criteria. Firstly, normally the vertebrae for BMD measurement using QCT are L1 and L2, which have lower BMD values than the thoracic vertebra, leading to false negative results when using the same cut-off values. Meanwhile, in our study, the lowest T value of lumbar spine and hip was used for BMD classification using DXA as the reference, while the AI software calculated BMD of the spine. It is possible that some patients had osteoporosis/osteopenia in the hip while the BMD of the spine was normal. Lastly, the results from DXA can be false positive due to positioning [24]. The low sensitivity indicated that the ACR osteoporosis diagnostic criteria for QCT cannot be applied directly when using the AI-derived BMD value of T11 for osteoporosis/osteopenia screening. According to the ROC analysis, the balanced cut-off value that yielded a highest Youden Index was 126 mg/cm³ for detection of osteoporosis vs osteopenia and normal subjects, and 163 mg/cm³ for detection of abnormal BMD vs normal subjects. For screening osteoporosis/osteopenia in high risk population, we would recommend higher cut-off values to achieve a 90 % sensitivity to avoid false negative results.

There are several limitations of our study. Firstly, the study was performed in a single institution. The conclusions from our study need to be further validated externally. The chest CT scans included in this study were acquired by two different CT scanners. Potential inter-device variation may exist. However, CT images acquired by multiple scanners in multiple institutions were used for development of this AI software and inter-device variation has been corrected. Thus we would not expect this inter-device variation to affect the conclusion of this study. Secondly, studies have shown that QCT can detect osteoporosis with higher sensitivity than DXA, and has a better correlation with osteoporotic vertebral fractures [25]. In our study, DXA results were used as the reference standard, which may suffer from false negative results caused by spinal degeneration and other overlapping structures. However, DXA is still the most widely used diagnostic tool for BMD analysis and treatment monitoring in osteoporosis. Thus the cut-off values derived from our study should be interpreted as the cut-off value for screening osteoporosis/osteopenia that can be diagnosed by DXA. In future, it is

necessary to perform further studies on the ability of this AI-derived BMD value of T11 to predict osteoporotic fractures, which can be more helpful for clinicians to interpret these BMD results.

In conclusion, the AI-derived BMD of T11 vertebra on chest CT scans has moderate to high diagnostic performance depending on cut-off thresholds for detection of osteoporosis and abnormal BMD using DXA results as the reference standard. The cut-off values for interpretation of the AI-derived BMD of T11 should be adjusted instead of directly using the ACR QCT diagnostic criteria for osteoporosis.

CRedit authorship contribution statement

Yongsheng Que: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xing Xia:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaozhen Luo:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Yaohong Deng:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Min Chen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical statement

This retrospective study was approved by the institutional ethical committee. Informed consent was waived.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ejrai.2025.100025](https://doi.org/10.1016/j.ejrai.2025.100025).

Data availability

Data will be made available on request.

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